P-S-625E
December 23, 1974
SUPERSEDING
P-S-625C
October 10, 1967, and
P-S-00625D (GSA-FSS)
October 29, 1970

FEDERAL SPECIFICATION

SOAP, BORAX (POWDERED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers powdered hand cleaner consisting of soap and borax (see 6.1).

1.2 Classification. Powdered borax soap shall be of the following classes, as specified (see 6.2 and 6.7).

Class 1 - Regular.
Class 2 - With lanolin.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.
P-S-620 - Soap, Toilet (Cake and Powdered).
FF-D-396 - Dispensers, Soap.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).
Fed. Test Method Std. No. 536 - Soap and Soap Products (Including Synthetic Detergents): Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

21 CFR 1 - Federal Food, Drug and Cosmetic Act.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Toilet Goods Association Inc. Specification:

No. 29 - Lanolin, Anhydrous.

(Application for copies should be addressed to the Toilet Goods Association, Inc., 9 Rockefeller Plaza, New York, NY 10020.)

American Society for Testing and Materials (ASTM) Standards:

D 460 - Sampling and Chemical Analysis of Soaps and Soap Products.
D 1172 - pH of Aqueous Solutions of Soaps and Detergents.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the product shall be commercial grade or superior. Lanolin shall be cosmetic grade and shall conform to Toilet Goods Association Specification No. 29.

3.1.1 Excluded material. The product shall contain no sugar or rosin.

3.2 Physical and chemical requirements. The product shall conform to the requirements of table I.

TABLE I. Physical and Chemical Requirements

Characteristic	Class 1		Class 2	
	min.	max.	min.	max.
Lanolin (percent)			2.5*	3.5
Matter insoluble in water (percent)	----	0.2	----	0.3
Anhydrous soda soap (percent)	23.0	27.0	23.0	39.0
pH	8.7	9.7	8.7	9.7
Borax, calculated as $Na_2B_4O_7.10H_2O$ (percent)	72.0	77.0	57.0	77.0

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TABLE I. Physical and Chemical Requirements (cont.)

Characteristic	Class 1 min.	Class 1 max.	Class 2 min.	Class 2 max.
Weight percent retained on:				
Number 12 U.S. Standard sieve	----	0.01	----	0.01
Number 40 U.S. Standard sieve	----	15.0	----	15.0
Number 100 U.S. Standard sieve	45.0	80.0	30.0	70.0

* If the weight percent of borax is in the range 72 to 77 percent, a minimum of 0.2 percent lanolin is permitted.

3.2.1 Computation. The test results shall be expressed as determined on the product as received. Since sodium tetraborate decahydrate ($Na_2B_4O_7.10H_2O$) effloresces, the sum of the percentages (of borax, anhydrous soap, and lanolin) determined on the product as received may exceed 100 percent. If this occurs, multiply the percentages (of borax, anhydrous soap, and lanolin) by the following factor:

$$\text{Multiplication factor} = \frac{100}{S}$$

where S = the sum of the percentages (of borax, anhydrous soap, and lanolin) as determined on the product as received.

3.3 Color. The product shall be uncolored or tinted in pastel shades of pink, green, or tan. If the product is dyed, the dye shall not impart a stain to the skin when the product is used. The manufacturer shall certify that the dyes used are certified by The Food and Drug Administration.

3.4 Odor. The product shall not have a rancid odor. The product may be mildly perfumed.

3.5 Performance in a dispenser. The product shall dispense freely and completely from a dispenser conforming to FF-D-396, type III. The product shall not clog or render inoperative any vital operating parts of the dispenser.

3.6 Freedom from abrasives. The product shall be free from grit and abrasive substances when tested as specified in 4.3.4.

3.7 Suds volume. The product shall produce not less than 100 milliliters of foam when tested as specified in 4.3.3.

3.8 Workmanship. The finished product shall be a clean, homogeneous and free-flowing powder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Certificates of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection.

4.2.1 Inspection of the end item. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase documents.

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4.2.1.1 Review of certificate of compliance. A certificate of compliance furnished by the manufacturer shall be reviewed to determine compliance of the lot with the requirement of 3.3.

4.2.2 Examination of preparation for delivery. Packaging, packing, and marking shall be examined to determine whether they comply with the requirements of section 5. Defects shall be scored as specified in table II. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-2, acceptable quality level (AQL) of 4.0 expressed in terms of defects per hundred units. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the lot (see 6.3 and 6.4).

TABLE II. Examination of preparation for delivery

Examine	Defect
Contents	Not as specified.
Containers	Not as specified.
Markings	Omitted; incorrect; illegible; improper size, location, sequence or method of application.
Materials	Component missing or damaged.
Workmanship	Bulging or distortion of containers. Cushioning inadequate, improper or missing.

4.2.2.1 Examination of closure, waterproofing and banding of shipping containers. When shipping containers are required to comply with PPP-B-636, examination for defects in closure, waterproofing and banding shall be in accordance with the appendix to that specification.

4.2.3 Examination of the end item. The end item shall be examined for defects in accordance with table III. Sampling and acceptance shall be in accordance with MIL-STD-105, inspection level S-4, AQL of 4.0 expressed in terms of defects per hundred units. The sample unit shall be one unit container (see 5.1). The lot size shall be the number of unit containers in the lot (see 6.3 and 6.4).

TABLE III. Examination of the end item

Examine	Defect
Net weight of container	Less than specified.
Contents of container	Any sifting of contents. Color not as specified. Rancid odor. Contents not clean and homogeneous.

4.2.4 Testing of the end item. The end item shall be tested as specified in table IV. Sampling shall be in accordance with MIL-STD-105, inspection level S-1. The sample unit shall be one pound of the product. The lot size shall be the number of pounds of product in the lot (see 6.3 and 6.4). Prepare a composite sample by thoroughly mixing the sample units selected as above. Draw a two-pound portion of the composite sample for testing. Place the two-pound portion into a clean, dry, moisture proof container and seal it. Label the container with the batch number, lot number, and the date. Perform two determinations of each test and report the average for numerical results. Failure of the composite sample to meet any of the requirements of table IV shall be cause for rejection of the lot.

Table IV. Instructions for testing the end item

Characteristic	Specification Reference		Results reported as	
	Requirement	Test Method	Pass or Fail	Average, Numerically to Nearest
pH	table I	4.3.2		0.1 pH unit
Matter insoluble in water	table I	ASTM D 460 #19		0.1 percent
Borax (calculated as $Na_2B_4O_7.10H_2O$)	table I	ASTM D 460 #53-55		0.1 percent

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TABLE IV. Instructions for testing the end item (cont.)

Characteristic	Specification Reference		Results reported as	
	Requirement	Test Method	Pass or Fail	Average, Numerically to Nearest
Lanolin	table I	4.3.1		0.1 percent
Anhydrous soap	table I	ASTM D 460 #21, 22		0.1 percent
Sieve tests	table I	2101*		0.1 percent
Suds volume	3.7	4.3.3		1.0 ml
Rosin	3.1.1	ASTM D 460 #37-40	X	
Sugar	3.1.1	1701*	X	
Abrasive	3.6	4.3.4	X	
Performance in a dispenser	3.5	4.3.5	X	
Stains skin on washing	3.3	4.3.6	X	

* Test method of Fed. Test Method Std. No. 536.

4.3 Test Methods.

4.3.1 Lanolin. The borax and soap are removed from a solution of the toilet soap in distilled water by a mixed cation-anion exchange resin; the resin is separated by filtration; the solvent is evaporated from the filtrate; and the residue weighed as lanolin.

4.3.1.1 Materials.

4.3.1.1.1 Ethanol-toluene azeotrope (ETA). Prepare by mixing 70 volumes of ethanol with 30 volumes of toluene.

4.3.1.1.2 Mixed cation-anion exchange resin. Rohm and Haas: "Amberlite Monobed MB-1", or eq 1. Test each batch to assure that it does not leave more than a small amount of ionic residue when used in the following procedure.

4.3.1.2 Procedure. Preset a thermostatically controlled hot plate, equipped with a magnetic stirrer, to maintain the temperature of the liquid contents of a beaker at 55°C ± 3°C. Weigh 5 grams (g) of the composite sample to the nearest milligram (mg). Place in a 250 ml beaker and add 100 ml distilled water. Heat on a water bath to 80°C, add stirring bar, and place on preset hot plate. Stir until all sample has been dissolved and temperature has dropped to 70°C. Add 85 g of cation-anion exchange resin and continue to stir for 35 minutes. Remove beaker from hot plate and allow the resin to settle. Filter the hot supernatent, with suction, through a sintered glass funnel (coarse porosity, 7 cm in diameter, 5 cm high) into a 500 ml filter flask, retaining as much resin as possible in the beaker.

Wash the resin by adding 30 ml of ETA to the beaker, washing down the sides of the beaker. Place the beaker on hot plate, stir and heat contents to 55°C. Remove beaker from hot plate, allow resin to settle, and filter the hot supernatent, with suction through the funnel, retaining as much resin as possible in the beaker.

Wash resin four more times as above. On the last wash, transfer entire contents of beaker to funnel. Rinse the beaker with 30 ml ETA and add to top of resin in funnel, distributing as uniformly as possible over the resin. After the rinse has been completely drawn through the funnel, transfer contents of filter flask to a tared (to nearest 0.1 mg) 400 ml beaker. Wash down the sides of the filter flask with a small portion of ETA. Heat flask and contents on hot plate until ETA vapor starts condensing near top of flask. Add rinse solution to filtrate. Repeat wash of filter flask. Evaporate filtrate on steam bath and cool beaker to room temperature.

Wash residue with 30 ml distilled water, pouring water directly on any ionic material in residue. Swirl water around the beaker and discard water. Repeat the washing two more times. Evaporate any water remaining in beaker on steam bath and dry beaker to constant weight dessicator. Calculate percent lanolin in sample as follows:

$$\text{percent lanolin} = \frac{A}{B} \times 100$$

where: A = weight of residue in grams.
B = weight of sample in grams.

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4.3.2 pH. Determine the pH in accordance with ASTM Method D 1172, except that a 1 percent solution by weight of the product shall be used.

4.3.3 Suds volume. Dissolve 2.5 ± 0.02 g of the composite sample in 250 ml of distilled water at 20°C ± 2°C. Place 50 ml of the solution in a 250 ml graduated cylinder. Stopper the cylinder and agitate by inversion, rotating the cylinder about its midpoint without translational motion for 1 minute at such rate that 60 inversions are completed. Place the cylinder upright on a table, remove the stopper, wait 5 seconds, and read the net volume of foam (total volume less volume of liquid).

4.3.4 Abrasives. Make a paste of 1 ml of distilled water and 1 g of the sample, mix well. Smear 1 g of the paste on a new glass microscope slide. Place a new glass microscope slide over the paste smear, press the slides together using moderate pressure; and maintaining pressure, rub the slides back and forth against each other fifty times. Wash the slides with distilled water and dry with a clean, soft cloth. Examine the slides in transmitted and reflected light without the use of a microscope. At least three people at the point of test shall examine the slides and agree that there are no scratches.

4.3.5 Performance in a dispenser. Fill a soap dispenser conforming to type III of FF-D-396 with 21.8 cubic inches (12 fluid ounces) of the composite sample. Operate the feed valve of the dispenser with full, uniform strokes until the composite sample is entirely removed. It shall take no less than 800 strokes and no more than 2,600 strokes to empty the dispenser. The composite sample shall not clog or render inoperative any vital operating parts of the dispenser.

4.3.6 Staining of the hands. Wash the hands with 1 gram of the composite sample and warm tap water. Rinse the hands well and dry with a clean, white towel. The hands shall not be stained from the dye in the product.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (see 6.2).

5.1.1 Level A. Five (5) pounds of powdered borax soap of one class only shall be packaged in a box conforming to PPP-B-566, variety 2, style 1, type A, class "b". The box shall be completely sealed with a permanent type of adhesive applied to the full area between the top and bottom flaps to prevent any sifting of the contents.

5.1.2 Level B. Five (5) pounds of powdered borax soap of one class only shall be packaged in a paperboard box conforming to PPP-B-566, variety 1, style 1, type A, class "b". The paperboard shall be double kraft lined chip calendared to a finish 2 of L-P-378. The paperboard shall have a minimum of 0.032 caliper and basis weight of 62 pounds. The box shall be completely sealed with an adhesive applied to the full area of the flaps to prevent any sifting of contents.

5.1.3 Level C. Powdered borax soap shall be packaged to afford adequate protection against loss or damage from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Ten five-pound boxes of powdered borax soap shall be packed in a box conforming to PPP-B-636, class weather resistant. The box shall be closed, waterproofed, and banded in accordance with the appendix of the box specification.

5.2.2 Level B. Ten five-pound boxes of powdered borax soap shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix of the box specification.

5.2.3 Level C. Powdered borax soap shall be packed to insure carrier acceptance and safe delivery from supply source to initial destination at the lowest rate for such suppliers. Containers shall comply with the requirements of the Uniform Freight Classification Rules or the National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.3.2 Military agencies. In addition to any special marking requirements by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Each shipping container shall be marked with the date of the manufacture of the soap powder.

6. NOTES

6.1 Intended use. The powdered borax soaps covered by this specification are intended to be dispensed from washroom dispensers conforming to type III of FF-D-396, for use with cold, soft water to remove heavy occupational soils such as oils and greases from the hands.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Special markings required (see 5.3.1 and 5.3.2).

6.3 Inspection lot. An inspection lot shall consist of all soap of the same class, manufactured in the same batch and offered for delivery at the same time.

6.4 Batch. A batch is defined as the quantity of material manufactured under the same processing conditions using the same raw material lots.

6.5 Basis of purchase. The basis of purchase for powdered borax soap shall be the shipping container.

6.6 Suggestions for improvement. Suggestions for improvement of this specification should be addressed to the General Services Administration, Federal Supply Service, FMBC, Washington, DC 20406.

6.7 Clas ification change. Type I under P-S-00625D (GSA-FSS), powdered toilet soap, has been transferred to P-S-620D, type II. Type II under P-S-00625D (GSA-FSS) is covered by this document.

Former classification	New classification
P-S-00625D (GSA-FSS), type I	P-S-620D, type II
P-S-00625D (GSA-FSS), type II, class 1	P-S-625E, class 1
P-S-00625D (GSA-FSS), type II, class 2	P-S-625E, class 2

MILITARY INTERESTS:

Army - GL (MCA)
Air Force - 84

CIVIL AGENCY COORDINATING ACTIVITIES:

VA - DMS
GSA - FSS
CPSC

Review Activity:

Army - MD

PREPARING ACTIVITY: GSA - FSS

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